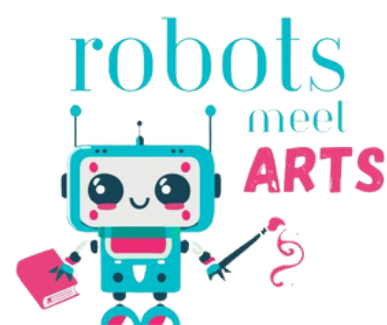




ROBOTS MEET ARTS

GUIDELINES FOR INCLUSIVE IMPLEMENTATION

WP3 - ROBOTS MEET ARTS CURRICULUM Guidelines for Inclusive Implementation

DATE

November 2025

RESPONSIBLE PARTNER

UCLL

COORDINATOR & PARTNERS



Co-funded by the European Union (Project number: 2023-1-FR01-KA220-SCH-000151881). Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or of the European Education and Culture Executive Agency (EACEA). Neither the European Union nor the granting authority can be held responsible for them.



Introduction

Implementing Inclusive Robotics and Coding Lessons

This guide accompanies the Robots Meet Arts Lesson Plans, a compilation of 29 ready-to-use activities for your classroom including: 8 introductory lessons (4 unplugged, 4 plugged) and 21 subject-specific lessons across humanities disciplines (Geography, Foreign Languages, Arts & Design, Civic & Moral Education, Mother Language, Music, and History). Each subject area includes three lessons: one unplugged activity, one using block-based coding, and one featuring a floor robot. While these lessons are designed to be comprehensive and engaging, this guide helps you adapt them to ensure every student in your classroom can participate fully and meaningfully.

Inclusive education in robotics and coding presents a significant challenge for primary school teachers. How can we ensure that all students – girls and boys, students with learning difficulties, children from diverse cultural backgrounds – can participate fully and develop their digital skills?

This practical guide addresses this question by synthesizing research-based best practices. It draws on the work of the Robots Meet Arts Training Module 4 “Development of inclusive and gender perspectives teaching material” and provides teachers with immediately applicable classroom strategies.



Why is inclusion in robotics and coding crucial?

Research demonstrates that STEM participation gaps begin in primary school. Social messages, classroom interactions, and pedagogical choices can either open or close access to technology for certain groups of students. These patterns intersect with learners' individual profiles — dyslexia, dyscalculia, autism, ADHD, different cultural backgrounds with language barriers — creating more obstacles than any child should face. The challenge is real: gaps start early and are reinforced by seemingly small moments in class, such as who holds the robot, who gets the follow-up question, whose ideas are heard first.

The encouraging reality: You can create inclusive STEM classrooms

As primary teachers, you can design and run inclusive robotics and coding lessons that work for every child. Universal Design for Learning (UDL) and Multisensory Pedagogy serve as your foundational approaches, helping you transform research into everyday practices through clear visuals and routines, multiple ways for students to demonstrate understanding, and fair assessment methods. In practice, this means creating predictable lesson flows, implementing role rotation so everyone gets hands-on time, offering options to respond through talking, drawing, using icons or alternative communication, and conducting quick comprehension checks that value both process and product.

Robotics and coding as transformative tools

At first glance, robotics and coding might seem like subjects that could reinforce existing inequities — expensive equipment that favors well-resourced students, technical complexity that might intimidate students who've been told "tech isn't for them," or abstract programming concepts that seem disconnected from diverse cultural experiences. However, when thoughtfully designed, these subjects possess unique characteristics that actually make them powerful tools for inclusion.

Robotics and coding are inherently hands-on, provide immediate feedback, and naturally encourage collaboration. When you design with inclusion from the start—using short, manageable coding steps, implementing "one change per run" protocols, providing visible timers for fair turn-taking, creating bilingual word banks, offering quiet sensory options—the activities themselves lower barriers. You diminish stereotypes while fostering growth mindsets. Abstract concepts become concrete: a loop becomes the robot repeating an action, a condition becomes the robot slowing at a crosswalk, a variable becomes the light threshold the team adjusts together.



Why is inclusion in robotics and coding crucial?

A proactive rather than corrective approach

Rather than retrofitting lessons after the fact, this guide emphasizes anticipating learner diversity from the planning stage. Inclusive pedagogy doesn't merely supplement robotics instruction—it fundamentally shapes the code structure, role assignments, and classroom routines. With deliberate pedagogical choices, your classroom becomes a space where every learner—girls, multilingual newcomers, and students with diverse learning profiles—can participate meaningfully and demonstrate measurable progress.

The goal: Psychologically safe, developmentally appropriate, academically rigorous

The objective is building classrooms that feel psychologically safe, match your students' developmental needs, and maintain rigorous STEM thinking. This balance is both achievable and essential for equitable technology education.



How to use this guide?

Part 1: Core principles and general checklists for daily design of inclusive activities

Use this when planning any robotics or coding lesson to ensure you're anticipating diverse learner needs from the start. The checklists help you avoid common exclusion patterns before they occur.

Part 2: Detailed strategies for addressing specific learner needs

Turn to this section when specific students aren't engaging or when you want to proactively support particular groups (girls, multilingual learners, students with learning differences). The comparative tables help you quickly identify appropriate interventions.

Part 3: Applying inclusive principles to Robots Meet Arts lesson examples

Use these concrete lesson adaptations as inspiration for your own teaching. See how the principles from Parts 1 and 2 are applied to specific robotics activities, with examples showing how to address different learner needs within the same lesson plan.



To deepen your knowledge

While the lesson plans and this guide can be used independently, the Robots Meet Arts project offers a complete training program that supports teachers educators in developing comprehensive competencies for implementing inclusive robotics and coding education in the field of humanities. This training, though not mandatory, provides valuable theoretical grounding and reflective distance that can significantly enhance your teaching practice.

The full program is available as a **self-paced online course on Moodle** and consists of six interconnected modules:

1. Introduction to ER and coding - Foundation concepts and principles
2. Teaching with ER and coding - Pedagogical approaches and classroom management
3. ER and coding activities in Arts and Humanities - Supporting the emergence of Lesson Plans
4. Developing Inclusive, Gender-Sensitive Teaching Materials - Theoretical foundation for inclusive practices
5. Familiarization with computational thinking and Educational Robotics - Technical tools and platforms
6. Knowledge put into practice & presentations - Assessment and implementation strategies

This structured progression allows you to build competencies systematically, from theoretical understanding to practical application, while developing critical reflection skills about their teaching practice.

This guide provides more specifically a practical synthesis of **Module 4 "Developing Inclusive, Gender-Sensitive Teaching Materials" from the "Robots Meet Arts" training program**. For comprehensive training including in-depth theoretical foundations, detailed case studies, and extended pedagogical resources, we strongly recommend consulting the complete Module 4, which offers:

- Fundamental pedagogical theories (Bronfenbrenner, CHAT, UDL)
- Several detailed pedagogical scenarios with all materials
- Assessment and self-reflection resources
- Complete bibliography and links to additional resources